Dividing (2) by (1) we get, $\frac{ar^6}{ar^3} = \frac{\frac{64}{243}}{\frac{8}{9}}$

$$r^3 = \frac{8}{27} \Rightarrow r = \frac{2}{3}$$

Substituting the value of r in (1), we get $a \times \left[\frac{2}{3}\right]^3 = \frac{8}{9} \Rightarrow a = 3$

Therefore the Geometric Progression is $a,\ ar,\ ar^2, \ldots$ That is, 3, 2, $\frac{4}{3}, \ldots$

Note

- When the product of three consecutive terms of a G.P. are given, we can take the three terms as $\frac{a}{r}, a, ar$.
- When the products of four consecutive terms are given for a G.P. then we can take the four terms as $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$.
- When each term of a Geometric Progression is multiplied or divided by a non–zero constant then the resulting sequence is also a Geometric Progression.

Example 2.44 The product of three consecutive terms of a Geometric Progression is 343 and their sum is $\frac{91}{3}$. Find the three terms.

Solution Since the product of 3 consecutive terms is given.

we can take them as $\frac{a}{r}, a, ar$.

Product of the terms $= 343$

$$\frac{a}{r} \times a \times ar = 343$$
$$a^3 = 7^3 \Rightarrow a = 7$$

Sum of the terms $= \frac{91}{3}$

Hence $a\left(\frac{1}{r} + 1 + r\right) = \frac{91}{3} \Rightarrow 7\left(\frac{1 + r + r^2}{r}\right) = \frac{91}{3}$

$3 + 3r + 3r^2 = 13r \Rightarrow 3r^2 - 10r + 3 = 0$

$(3r - 1)(r - 3) = 0 \Rightarrow r = 3$ or $r = \frac{1}{3}$

If $a = 7$, $r = 3$ then the three terms are $\frac{7}{3}, 7, 21$.

If $a = 7$, $r = \frac{1}{3}$ then the three terms are $21, 7, \frac{7}{3}$.

Thinking Corner

1. Split 64 into three parts such that the numbers are in G.P.
2. If $a, b, c, \ldots$ are in G.P. then $2a, 2b, 2c, \ldots$ are in _____
3. If 3, x, 6.75 are in G.P. then x is _____

Progress Check

Three non-zero numbers a, b, c are in G.P. if and only if _____.

Condition for three numbers to be in G.P.

If a, b, c are in G.P. then $b = ar$, $c = ar^2$. So $ac = a \times ar^2 = (ar)^2 = b^2$. Thus $b^2 = ac$

Similarly, if $b^2 = ac$, then $\frac{b}{a} = \frac{c}{b}$. So a, b, c are in G.P.

Thus three non-zero numbers a, b, c are in G.P. if and only if $b^2 = ac$.

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